

First-Stage Research Note

Thermal Extremes and Cardiovascular Admissions in Hong Kong, 2013–2023

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This note distinguishes real public-data findings from synthetic workflow testing. No real Hospital Authority outcome association has yet been estimated.

Executive Summary

During the first stage of the project, I built and executed a reproducible R workflow intended to support the eventual analysis of monthly Hospital Authority (HA) cardiovascular admissions. I also processed real Hong Kong Observatory (HKO) weather data into monthly averages and officially defined thermal-extreme measures.

The principal findings at this stage are methodological and descriptive:

- The HKO exposure-processing pipeline reproduces published annual counts of hot nights, very hot days, and cold days for all 11 years 2013–2023 (33/33 annual totals match HKO *Year’s Weather* summaries).
- The study period contains substantial variation in thermal extremes. For example, HKO recorded 10 hot nights and 17 very hot days in 2013, compared with 61 hot nights and 54 very hot days in 2021.
- Cold exposure has not disappeared. HKO recorded 14 cold days in 2013 and 13 in 2021. The appropriate scientific question is therefore not whether heat has simply replaced cold, but whether an emerging heat burden now coexists with persistent cold-related risk.
- The planned health-outcome workflow has been tested using synthetic counts. This validates the mechanics of data merging, offsets, count models, diagnostics, and output generation, but provides no evidence about actual AMI or stroke admissions.
- The primary remaining dependency is a clearly defined, securely accessible HA outcome extract. Public C&SD population and EPD pollution data can be completed in parallel.

The next scientific milestone is therefore not a more elaborate model. It is agreement on the HA outcome definition, episode structure, age bands, and secure data-access process.

1 Research Question and Current Framing

Earlier Hong Kong research used daily observations and reported a strong association between cooler temperatures and acute myocardial infarction (AMI) admissions. Goggins et al. found that, below an

estimated threshold of 24°C, each 1°C decrease was associated with a 3.7% increase in Hong Kong AMI admissions using an average temperature lag of 0–13 days. No statistically significant heat effect was detected, and same-day nitrogen dioxide was the strongest pollutant predictor [1].

The present project asks whether the contemporary cardiovascular risk profile remains predominantly cold-related or now includes detectable associations with increasingly frequent heat extremes. The proposed study population is Hong Kong residents aged 35 years and older, and the intended outcomes are monthly admissions for AMI, ischemic stroke, and hemorrhagic stroke.

The most defensible current framing is:

How are officially defined thermal extremes—particularly hot nights and cold days—associated with contemporary cardiovascular hospital burden in an aging Hong Kong, and how do these associations vary by diagnosis and age?

This framing does not assume in advance that a “regime shift” has occurred. A persistent cold-dominant association would remain scientifically and operationally important.

2 Work Completed

2.1 Climate-data processing

I built the weather-processing portion of the R pipeline using HKO daily observations at the Observatory Headquarters. It converts daily data into monthly variables including mean, minimum, and maximum temperature; relative humidity and rainfall; hot nights ($T_{\min} \geq 28^\circ\text{C}$); very hot days ($T_{\max} \geq 33^\circ\text{C}$); extremely hot days ($T_{\max} \geq 35^\circ\text{C}$); cold days ($T_{\min} \leq 12^\circ\text{C}$); and longest consecutive runs of hot nights and very hot days.

The pipeline includes date coverage, missingness, plausible-value, and annual-total checks. Annual extreme counts for 2013–2023 agree with the corresponding published HKO summaries (see Figure 1 and the validation table in the repository).

2.2 Statistical dry run

I created an explicitly synthetic HA-like panel with structure $\text{month} \times \text{age} \times \text{sex} \times \text{diagnosis}$ to test merging, $\text{population} \times \text{days}$ offsets, Poisson/Negative Binomial specifications, interactions, COVID sensitivities, diagnostics, and automatic table/figure generation. This demonstrates that the workflow can accept an HA file once its structure is finalized. It does **not** produce substantive cardiovascular findings.

2.3 Research documents

The repository also contains draft materials for the analysis plan and assumption log; data requests for Roro and Hogan; a preliminary Methods section; table and figure shells; and a literature baseline. The purpose of the present note is to summarize the scientifically relevant content without requiring the reader to inspect the code.

3 Preliminary Findings from Real HKO Data

Table 1 presents selected benchmark years from official HKO annual summaries. It is not intended as a formal trend analysis.

Table 1: Selected HKO thermal-extreme benchmarks					
Year	Mean temp. (°C)	Hot nights	Very hot days	Cold days	Status
2013	23.3	10	17	14	Core period
2017	23.9	41	29	9	Core period
2021	24.6	61	54	13	Core period
2024	24.8	50	52	11	Context only

1. The study period contains meaningful heat-exposure contrast

The later study period contains substantially more hot nights and very hot days than 2013. This is useful analytically: the 2013–2023 period is not uniformly hot and therefore contains exposure variation with which to estimate associations. This should not yet be described as a formal trend finding.

2. Hot nights are a defensible primary heat metric

Monthly average temperature can obscure short episodes of extreme exposure. Hot nights measure the absence of overnight thermal relief and are defined consistently by HKO. Very hot days, mean minimum temperature, heat-spell duration, and wet-bulb temperature should be considered alternative exposure specifications rather than entered simultaneously into one highly collinear primary model.

3. Cold remains part of the exposure distribution

Cold days remained present even in warm years. The project should therefore retain both heat and cold metrics and avoid framing the analysis as a forced competition in which one must replace the other.

4 Lessons from the Coding and Dry Run

4.1 Population offsets are essential

Because the older population changed substantially during the study period, raw admission counts cannot be interpreted as changing risk. The proposed offset is $\log(P_{ast} \times D_t)$, where P_{ast} is the population for age group a , sex s , and month t , and D_t is days in that month.

4.2 The effective environmental sample is 132 months

Although stratification may produce several thousand outcome rows, the territory-wide environmental exposures vary across only 132 months. Inference will need to recognize common monthly exposure and repeated observations within outcome strata.

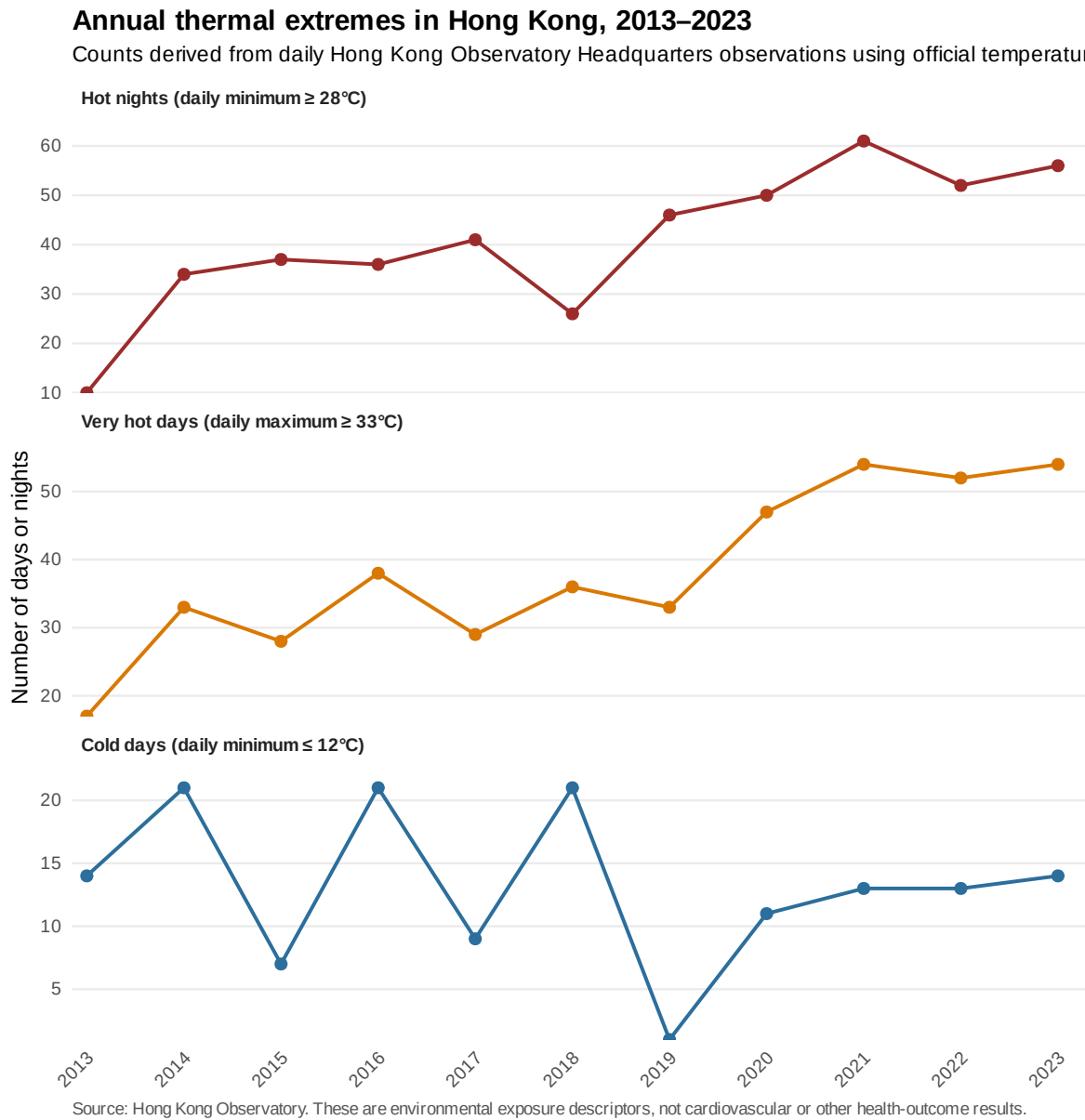


Figure 1: Annual counts of hot nights, very hot days, and cold days at the Hong Kong Observatory Headquarters, 2013–2023. Hot nights are defined as days with a minimum temperature of at least 28°C ; very hot days as days with a maximum temperature of at least 33°C ; and cold days as days with a minimum temperature of at most 12°C . These are environmental exposure descriptors and not health-outcome results. Pipeline annual totals match official HKO *Year's Weather* summaries for all years shown (33/33 checks).

4.3 Monthly data favor burden estimation over acute-trigger analysis

Earlier Hong Kong studies used daily data and distributed lag structures. Heat effects may occur over short lags, while cold effects may persist over longer periods. Monthly aggregation may therefore attenuate heat associations more strongly than cold associations. Results should be interpreted as monthly hospital-burden associations, not estimates of same-day physiological triggering.

4.4 Outcome definition matters as much as model choice

Before analysis, it is necessary to determine whether the file counts admissions, patients, or completed episodes; whether principal diagnosis is available; how transfers and readmissions are handled; whether ED attendance followed by admission is counted twice; and whether emergency and elective admissions can be distinguished.

5 Proposed Statistical Framework

The primary analyses should be diagnosis-specific. A provisional Negative Binomial model includes heat exposure H_t (initially monthly hot nights), cold exposure C_t , age, sex, diagnosis, month effects, a smooth time trend, prespecified covariates $\mathbf{Z}_t$, and the population–time offset. Poisson regression should be fitted as a baseline and checked for overdispersion. Principal effect estimates would be rate ratios per five additional hot nights or cold days. Interactions (thermal $\times$ age, thermal $\times$ diagnosis, heat $\times$ ozone) should be introduced separately.

6 Data Status and Transfer

Public environmental and population files can ordinarily be exchanged through institutional email or a shared drive. No patient-level, restricted, small-cell, or unvetted HA output should be sent by ordinary email until Roro confirms governance status.

7 Questions That Require Team Input

Four issues are immediate blockers:

1. **Coding and diagnosis position.** ICD-9-CM, ICD-10, or local coding? Can principal diagnoses be identified?
2. **Unit of outcome.** Admissions, unique patients, or completed episodes? How are transfers and readmissions handled?
3. **Granularity and suppression.** Can five-year age bands be supplied, especially 65–69 and 70–74? What small-cell rules apply?
4. **Secure access.** Secure-server analysis versus approved aggregate transfer? Which packages and output-vetting procedures apply?

Choices that can be deferred without delaying the project include extending outcomes through 2024, district analyses, comorbidity extensions, the final pollution station set, influenza in primary versus

Table 2: Current data status

Source	Status	What is needed	Proposed handling
HKO weather	Real data processed	Complete final quality review; add wet-bulb or multi-station measures only if they improve the prespecified analysis.	Public data; institutional email, repository, or shared drive.
C&SD population	Pending	Download the half-yearly age-sex population table and construct monthly denominators, retaining 65–69 and 70–74 where possible.	Public CSV/XLSX/JSON; shared drive or repository.
EPD pollution	Pending	Download NO ₂ , O ₃ , PM _{2.5} , and PM ₁₀ for selected general stations (monthly portal limit implies ≥ 2 exports for 132 months).	Public CSV; shared drive or institutional email.
HA outcomes	Pending	Confirm coding, principal diagnosis, episode structure, age bands, suppression, and approved analysis environment.	Approved HA/HKU secure process only unless Roro confirms otherwise.
Synthetic outcomes	Synthetic only	Retain only for pipeline tests; never merge into substantive output.	Repository, clearly labelled synthetic.

sensitivity models, and the exact correlated-count model for final inference. My current default is to retain 2013–2023 as the core period and request 2024 only if readily available.

8 Proposed Next Steps

1. Send Roro focused HA extraction questions and request a mock schema or data dictionary.
2. Download and process real C&SD population data.
3. Obtain EPD files for NO₂, O₃, PM_{2.5}, and PM₁₀ (general stations first).
4. Produce a final environmental QC report (missingness, correlations, annual exposure plots).
5. Freeze a short pre-analysis plan after the HA schema is confirmed.
6. Run descriptive validation of HA counts before regression.
7. Fit the prespecified model ladder.

I have also begun a structured literature update, which I will use to refine the Introduction and prespecify historical comparisons before examining HA results.

9 Proposed Meeting

A 30-minute meeting with Dr. Bishai, Hogan, Roro, and Bob would be sufficient to unlock the next stage: confirm the primary question and 2013–2023 core period; confirm HA outcome, coding, and episode

structure; confirm age bands and suppression; confirm secure access; allocate C&SD/EPD responsibility; and agree a date for the first validated HA descriptive report.

Conclusion

The first-stage work has established that the environmental exposure pipeline is feasible, reproducible, and capable of representing both heat and cold extremes. It has also produced an analysis framework ready to receive the clinical and population files. The project has not yet produced a cardiovascular association result. The next step should prioritize outcome validity and secure data integration rather than additional model complexity.

A Working Manuscript Introduction

Temperature-related cardiovascular morbidity is an important concern in subtropical cities, where populations may experience both intense humid heat and episodic cold despite generally warm climates. Acute myocardial infarction and stroke may respond differently to thermal exposure, and the observed burden can vary according to age, housing, behavioral adaptation, air pollution, and access to temperature-controlled environments. These questions are especially relevant in Hong Kong, a densely populated and rapidly aging city with substantial seasonal variation and a growing frequency of hot nights and very hot days.

Earlier Hong Kong evidence emphasized cold-related cardiovascular risk. Using daily data from 2000–2009, Goggins et al. reported that decreases in temperature below an estimated 24°C threshold were associated with increased AMI admissions, while no statistically significant heat effect was observed [1]. The same study identified nitrogen dioxide as an important predictor of AMI admissions.

Hong Kong’s thermal environment has changed since these historical study periods. At the Hong Kong Observatory, the number of hot nights increased from 10 in 2013 to 61 in 2021, while very hot days increased from 17 to 54 in the same selected-year comparison [2, 4]. Cold days nevertheless continued to occur, indicating that contemporary preparedness must consider potentially overlapping heat and cold hazards.

Demographic change may further alter the distribution of cardiovascular burden. Age-sex-specific population denominators and appropriate person-time offsets are necessary to distinguish changing population structure from changes in admission rates. The environmental context has also evolved: temperature and pollutants should not be treated as statistically interchangeable covariates, and ozone in particular may act as a confounder, effect modifier, or pathway variable.

This study will examine monthly AMI, ischemic stroke, and hemorrhagic stroke admissions among Hong Kong residents aged 35 years or older from 2013 to 2023. It will assess associations with HKO-defined thermal extremes, emphasizing hot nights and cold days, while accounting for population denominators, seasonality, long-term trends, and COVID-period disruption. The objective is not to assume that heat has replaced cold, but to determine whether the historically cold-dominant pattern remains sufficient to describe contemporary cardiovascular hospital burden in a warming and aging Hong Kong.

B Use of AI-Assisted Tools

AI-assisted tools were used for code scaffolding, methodological brainstorming, and review of potential data and modeling issues. Synthetic data were used to test the workflow before access to HA outcomes. All substantive claims about HKO annual extremes were checked against official HKO *Year's Weather* pages. Literature citations used in this note were restricted to verified primary sources; a fuller evidence matrix is under construction as an internal deliverable.

References

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